Submission Template README

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1. Introduction

This section will give a detailed description of the .xlsx file submission template structure as well as instructions on how to populate it.

1.1 Instructions

A .xlsx file for each specific scenario (N) iteration (M) is provided with the file naming convention reported below. Teams must submit one .xlsx file per scenario and iteration in the evaluation folder keeping the original filenames.

scenario_<N>_iteration_<M>.xlsx

In each .xlsx file, the `timestamp` and `unix_timestamp` columns are pre-filled at 100 milliseconds [ms] intervals, spanning from the first to the last ground-truth timestamp of that specific iteration.

Teams' task is that of filling in the remaining columns with their sensor-fusion algorithm's output. Participants must follow four main filling rules:

I. For each timestamps where the tracker maintains a track, add one row per tracked target;
II. For each timestamps where the algorithm has no active tracks, leave the corresponding row as-is (empty row).
III. If the algorithm tracks K>1 objects at a given timestamp, add K rows sharing that timestamp — one per target;

IV. If the algorithm can't estimate the values of a column, fill the corresponding cell with the string "Unknown" and/or chose the "Unknown" option where a dropdown menu is present;

V. If the algorithm value of a column can't be estimated by the algorithm due to lack of data, fill the corresponding cell with the string "Not applicable" and/or chose the "Not applicable" option where a dropdown menu is present;

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1.2 .xlsx file structure

The submission' .xlsx file column structure is reported in Table C.1. Detailed information on the column entries can be found at the following paragraphs.

Table C.1 The submission .xlsx file column structure

	#	Column	Type	Unit/Format	Description
	1	`timestamp`	string	ISO 8601 (µs precision), UTC	Do not modify
	2	`unix_timestamp`	float	milliseconds since Unix epoch, UTC	Do not modify
	3	`latitude`	float	degrees (WGS 84)	Component of fused position
	4	`longitude`	float	degrees (WGS 84)	Component of fused position
	5	`altitude_m`	float	metres (Height above the WGS84 ellipsoid)	Component of fused position
	6	`C_nn`	float	m^2	Covariance: North–North
	7	`C_ne`	float	m^2	Covariance: North–East
	8	`C_nu`	float	m^2	Covariance: North–Up
	9	`C_ee`	float	m^2	Covariance: East–East
	10	`C_eu`	float	m^2	Covariance: East–Up
	11	`C_uu`	float	m^2	Covariance: Up–Up
	12	`track_id`	GUID/ UUID	-	Unique ID per tracked target from your algorithm
	13	`classification`	Set of strings	-	Object type
	14	`classification_confidence`	float	0.0 – 1.0	Classification confidence
	15	`identity`	Set of strings	-	Target identification
	16	`uav_model/manufacturer`	Set of Strings	-	Target model and/or manufacturer (if applicable)
	17	`protocol`	Set of strings	-	Target communication protocol (if applicable)
	18	`serial_number`	Set of strings	-	Target serial number (if applicable)
	19	`signal_frequency_band`	Set of strings	MHz	Target signal frequency (if applicable)

1.2.1 State vector

Teams' result should report the state vector as a 3D position in WGS 84 geodetic coordinates as expressed in (C.1).

$$x = [latitude\ (°), longitude\ (°), altitude\ (m)] \quad \text{(C.1)}$$

1.2.2 Covariance matrix

Teams' result should report the upper triangle of the covariance matrix C of the state vector to express their tracker positional uncertainty along each axis. The proposed sensor-fusion algorithm should measure the covariance matrix with a local Cartesian system centred on the state vector position and oriented North-East-Up (NEU). Covariance matrix expression can be found in (C.2).

$$C = \begin{array}{c} \\ N \\ E \\ U \end{array} \begin{array}{c} \begin{array}{ccc} N & E & U \end{array} \\ \begin{bmatrix} c_{nn} & c_{ne} & c_{nu} \\ & c_{ee} & c_{eu} \\ & & c_{uu} \end{bmatrix} \end{array} \quad \text{(C.2)}$$

Where:

- $C \epsilon \mathbb{R}^{3x3}$ represent the covariance matrix;
- $c_{nn}, c_{ee}, c_{uu} \epsilon [0, +inf)$ represent variances in m²;
- $c_{ne}, c_{nu}, c_{eu} \epsilon [-inf, +inf)$ represent covariances m².

Positional uncertainty expressed in WGS 84 geodetic coordinates should be converted in positional uncertainty expressed in the North-East-Up (NEU) coordinate system by using the Jacobian as reported in (C.3).

$$J = J_{ECEFtoNEU} \cdot J_{GEOtoECEF} \quad \text{(C.3)}$$

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$$J_{GEOtoECEF} = \begin{bmatrix} \frac{\partial X}{\partial \varphi} & \frac{\partial X}{\partial \vartheta} & \frac{\partial X}{\partial h} \\ \frac{\partial Y}{\partial \varphi} & \frac{\partial Y}{\partial \vartheta} & \frac{\partial Y}{\partial h} \\ \frac{\partial Z}{\partial \varphi} & \frac{\partial Z}{\partial \vartheta} & \frac{\partial Z}{\partial h} \end{bmatrix},$$

$$J_{ECEFtoNEU} = \begin{bmatrix} -\sin(\varphi_0)\cos(\vartheta_0) & -\sin(\vartheta_0) & \cos(\varphi_0)\cos(\vartheta_0) \\ -\sin(\varphi_0)\sin(\vartheta_0) & \cos(\vartheta_0) & \cos(\varphi_0)\sin(\vartheta_0) \\ \cos(\varphi_0) & 0 & \sin(\varphi_0) \end{bmatrix}$$

Where

- $J_{GEOtoECEF} \in \mathbb{R}^{3x3}$, which represents the Jacobian of the transformation from the geodetic coordinates system (φ, ϑ, h) to the ECEF, Earth-Centered-Earth-Fixed coordinate system (X,Y,Z);
- $J_{ECEFtoNEU} \in \mathbb{R}^{3x3}$, which represents the rotation matrix from the ECEF coordinate system (X,Y,Z) to the North-East-Up (NEU) coordinate system (n,e,u);
- (φ_0, ϑ_0), which represents the reference point latitude and longitude respectively in [rad].

1.2.3 Track id

Teams' result should assign a track identification number as UUID/GUUID, unique per each object tracked by their proposed sensor-fusion algorithm. The same track number must be assigned at the same target across all timestamps of the corresponding scenario and iteration.

1.2.4 Classification

Teams' result should choose from the corresponding dropdown menu one of the strings listed as "Value" in Table C.2 to specify the object type tracked by their proposed sensor-fusion algorithm.

Teams' result should report classification confidence information as s unique value $\rho\epsilon[0,1]$ per each object type classified by their proposed sensor-fusion algorithm.

Table C.2 Object types

Column	Value	Meaning
`classification`	`Fixed Wing UAS`	Fixed-wing unmanned aerial system
	`Rotary Wing UAS`	Multi-rotor / rotary-wing UAS
	`UAS Transmitter`	RF transmitter associated with a UAS
	`Unknown`	Not able to determine type
	`Not Applicable`	Lack of data to determine type

1.2.5 Identification

Teams' result should choose from the corresponding dropdown menu one of the strings listed as "Value" in Table C.3 to identify the object tracked by their proposed sensor-fusion algorithm. Values has been chosen in accordance with NATO Standard 1241.

Table C.3 Object Identity

Column	Value	Meaning
`identity`	`Hostile`	Hostile
	`Friendly`	Friendly
	`Unknown`	Not able to determine identity
	`Not Applicable`	Lack of data to determine identity

1.2.6 Class metadata

Teams' result should choose from the corresponding dropdown menu one of the strings listed as "Value" in Table C.4 to specify the class/identity of the object tracked by their proposed sensor-fusion algorithm.

Table C.4 Object class specification

Column	Values	Meaning
`uav_model` / `manufacturer`	DJI Mavic 3 DJI Mavic 3 Enterprise MTRC 300R Autel Drone Parrot Anafi FPV FPV Chimera Unknown Not Applicable	Target model and/or manufacturer
`protocol`	DJI Ocusyinc Autel Skylink Wifi Video Digital video link type 2 DIY Crossfire Unknown Not Applicable	Target communication protocol
`serial_number`	F45TB21BR1AE03FP 163DG1700154R3 F5FHD233M00DEYG6 1ZNBJAB00C00MQ 1748FEV2HL7824051124 Unknown Not Applicable	Target serial number / unique ID number
`signal_frequency_band`	2400 – 2483.5 5725 – 5850 902 – 928 5150 – 5250 2200 – 2500 Unknown Not Applicable	Target signal frequency band [MHz]

1.2.7 Multiple targets per timestamp

Teams' result should report one row per each K>1 tracks sharing the same timestamp as per the following example.

```xlsx

timestamp,latitude,longitude,altitude_m,C_nn,C_ne,C_nu,C_ee,C_eu,C_uu,classification, classification_confidence, identity, uav_model/manufacturer, protocol, serial_number, signal_frequency_band, node_id, object-id

2024-09-16T07:27:00.000000,51.525131,5.869111,72.0,1e-8,0,0,1e-8,0,9.0, Fixed Wing UAS, 0.8, Hostile, DJI Mavic 3, DJI Ocusyinc , Not Applicable, 2400 – 2483.5
```

2024-09-16T07:27:00.000000,51.511386,5.837389,80.5,1e-8,0,0,1e-8,0,4.0, Rotary Wing UAS, 0.75, Friendly, Unknown, Wifi, Not Applicable, 5725 – 5850

```
```